Laboratory 2
Grid Search for Linear Regression

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Overview of exercises

- Laboratory 1: K-Means Clustering (1.0 Point)
- Laboratory 2: Grid Search for Linear Regression (1.0 Point)
- Laboratory 3: Gradient Descent for Linear Regression (1.5 Point)
- Laboratory 4: Logistic Regression (1.0 Point)
- Laboratory 5: Basics of Neural Network (1.0 Point)
- Laboratory 6: Image Classification with Neural Network (1.5 Points)
- Laboratory 7: Reinforcement learning (1.5 Point)
- Laboratory 8: Deep RL (1.5 Point)

Example: Dataset

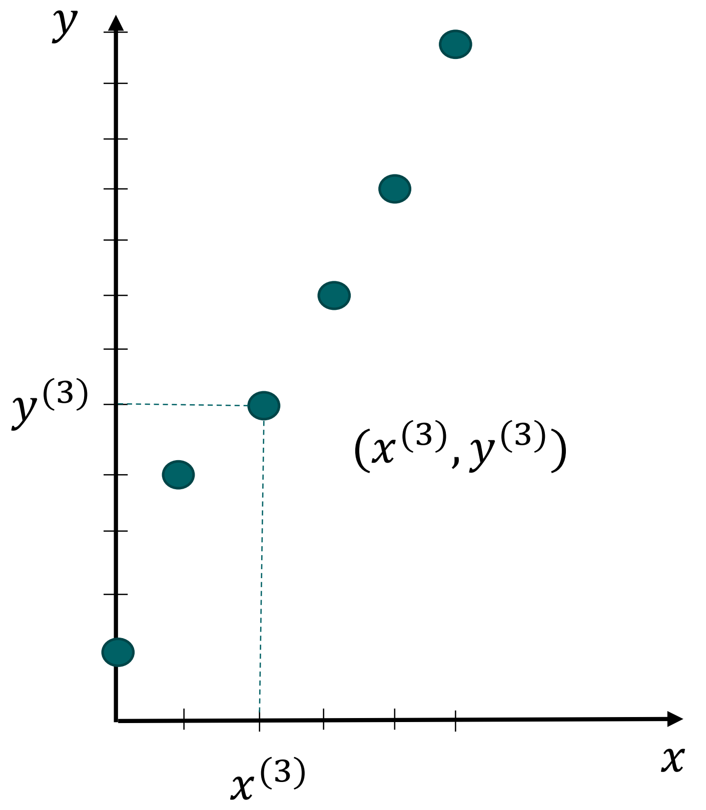
- Dataset consists of 6 examples and we randomly choose 4 examples as training set and 2 examples as testing set.

Dataset $\{(x^{(i)}, y^{(i)})\}_{i=1}^{6}$

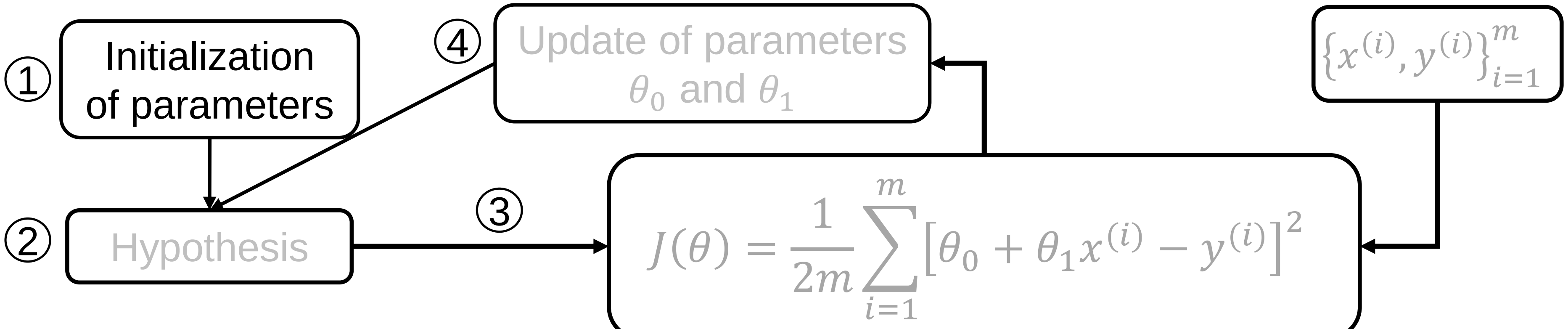
	point	value
1	$(x^{(1)}, y^{(1)})$	(0,1)
2	$(x^{(2)}, y^{(2)})$	(1,4)
3	$(x^{(3)}, y^{(3)})$	(2,5)
4	$(x^{(4)}, y^{(4)})$	(4,9)
5	$(x^{(5)}, y^{(5)})$	(5,12)
6	$(x^{(6)}, y^{(6)})$	(3,7)

Training set

Testing set



Linear Regression Training- ① Initialization



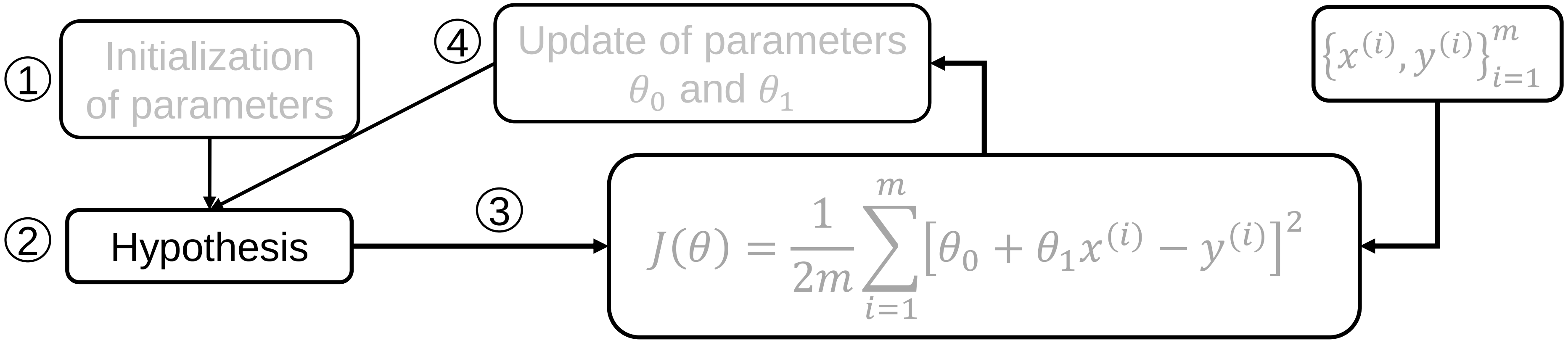
1 Parameters Initialization

- Let‘s assume $\boldsymbol{\theta} = [\theta_0, \theta_1]$ is defined by

$$\theta_0 = 0$$

$$\theta_1 = 1$$

Linear Regression Training- ② Hypothesis



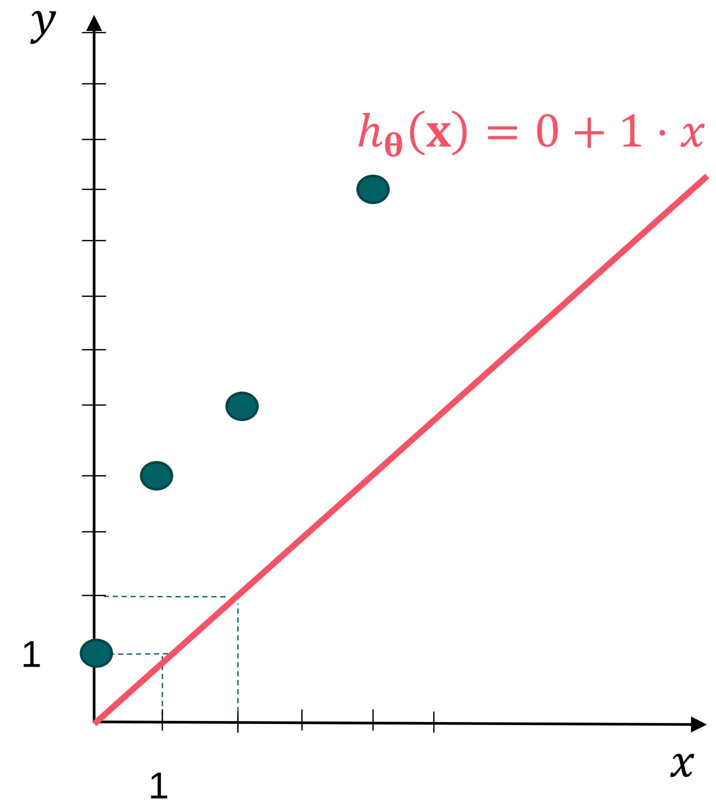
② Hypothesis function

- The linear regression model is:

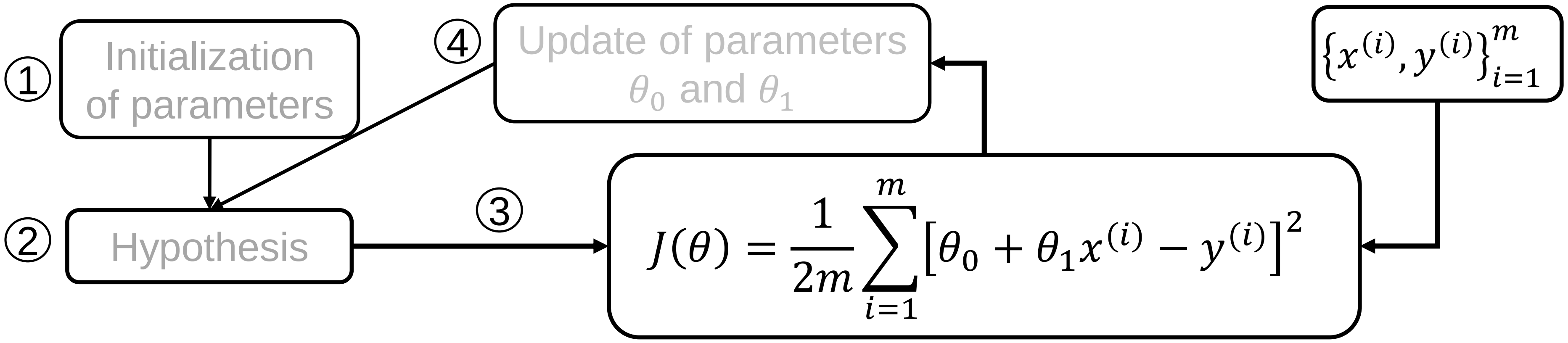
$$h_{\boldsymbol{\theta}}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$$

- For the given $\boldsymbol{\theta}$, the model is:

$$h_{\boldsymbol{\theta}}\left(x^{(i)}\right) = 0 + 1 \cdot x^{(i)}$$



Linear Regression Training- 3 Cost function

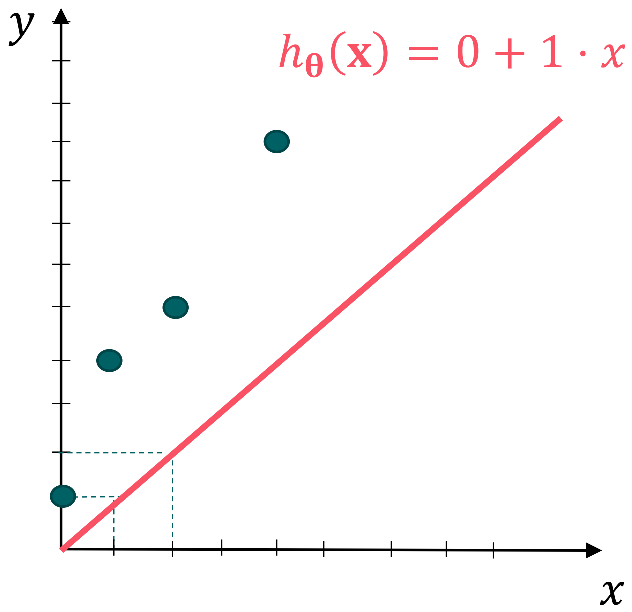


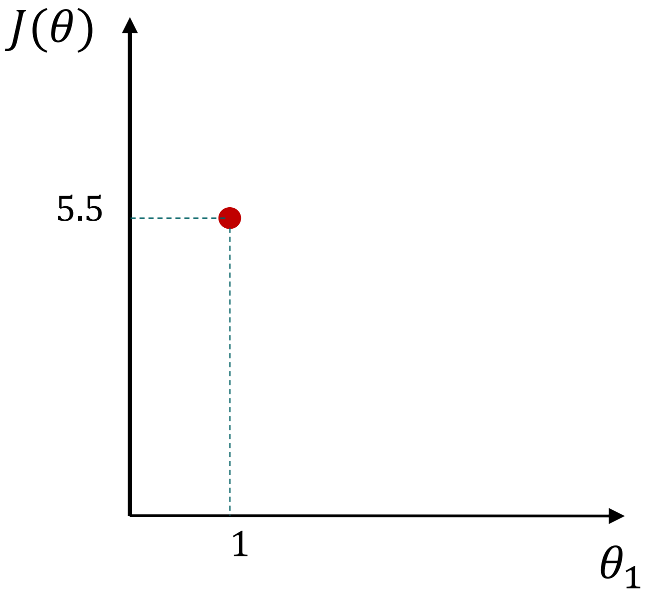
③ Cost function computation $J(\theta_0 = 0, \theta_1 = 1)$

$$J(\boldsymbol{\theta}) = \frac{1}{2m}\sum_{i=1}^{m=4}\left[h_{\boldsymbol{\theta}}\left(x^{(i)}\right) - y^{(i)}\right]^2$$

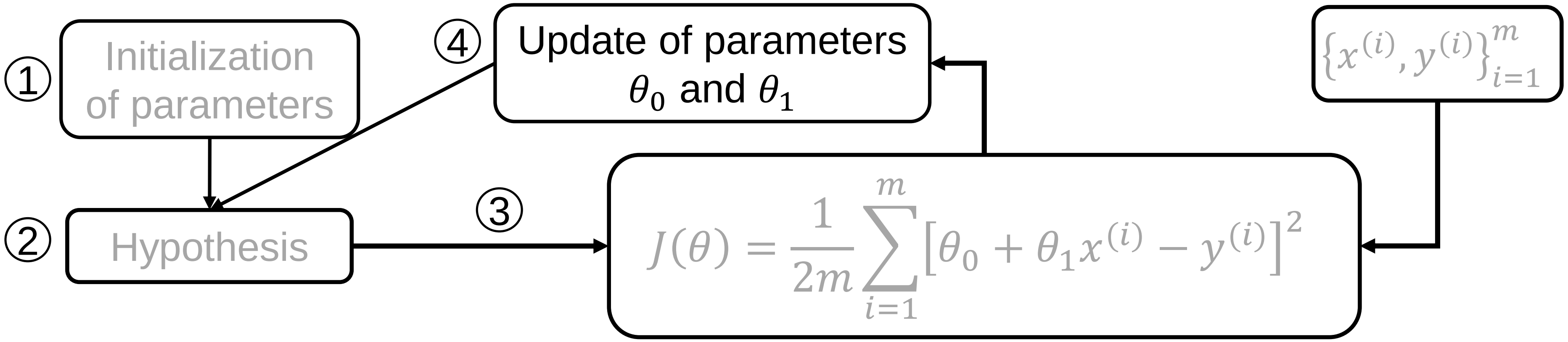
$$= \frac{1}{2m}\sum_{i=1}^{m=4}[0 + 1 \cdot x^{(i)} - y^{(i)}]^2$$

$$= \frac{1}{2 \times 4}([0-1]^2 + [1-4]^2 + [2-5]^2 + [4-9]^2) = 5.5$$





Linear Regression Training- ④ Parameter update



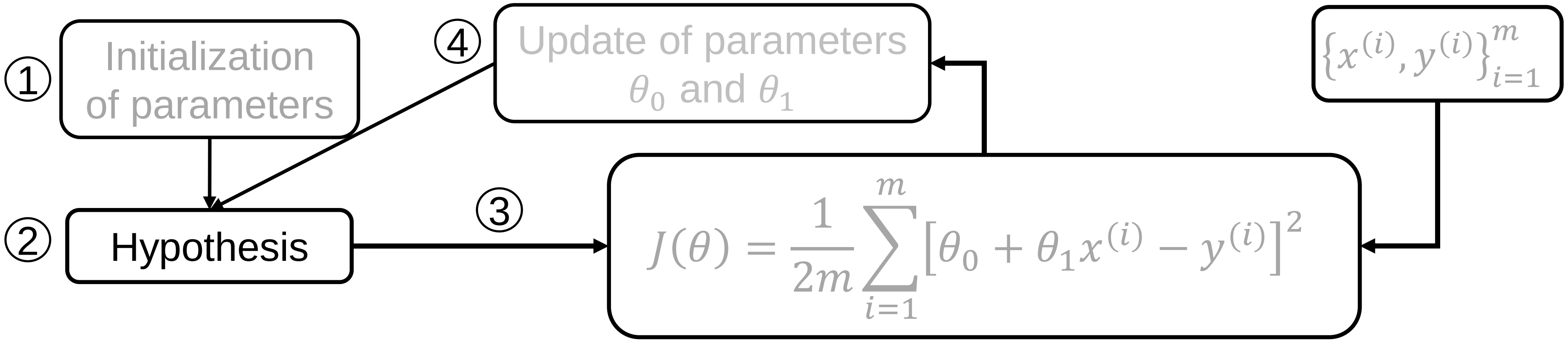
4 Parameters Update for first iteration

- Let‘s assume the updated parameters are as follows:

$$\theta_0 = 0$$

$$\theta_1 = 2$$

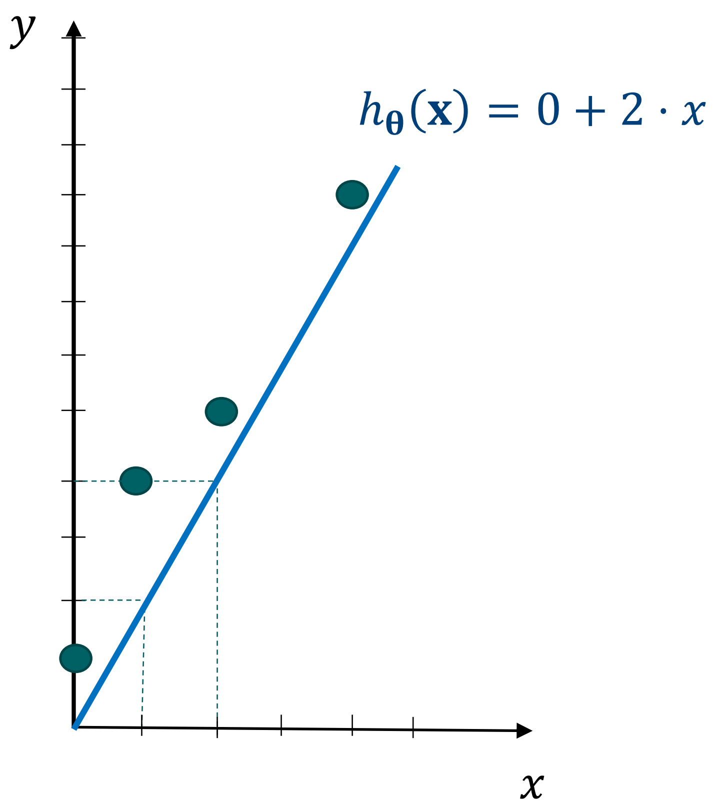
Linear Regression Training- 2' Hypothesis



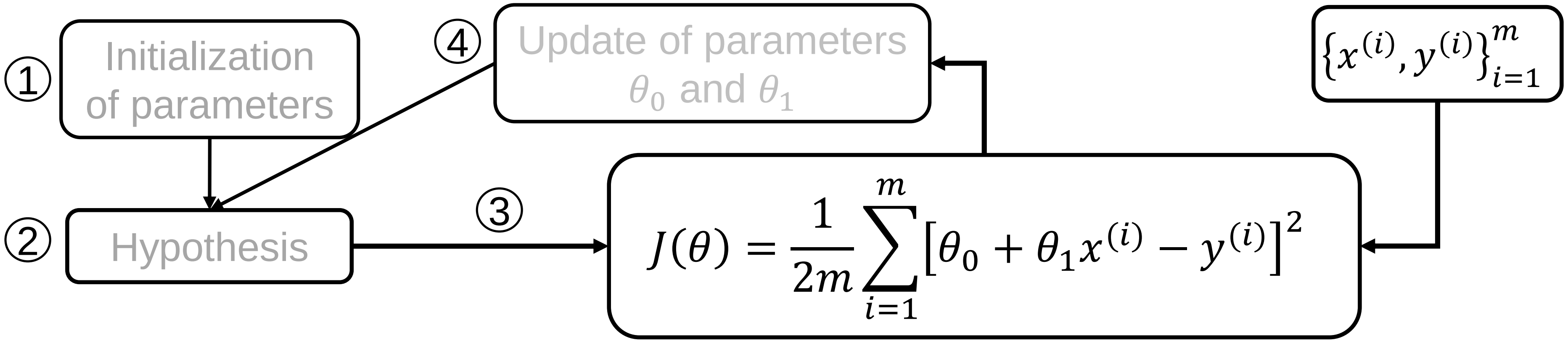
2' Hypothesis for first iteration

- Then, for the given $\boldsymbol{\theta}$, the model is:

$$h_{\boldsymbol{\theta}}\left(x^{(i)}\right) = 0 + 2 \cdot x^{(i)}$$



Linear Regression Training- 3' Cost function

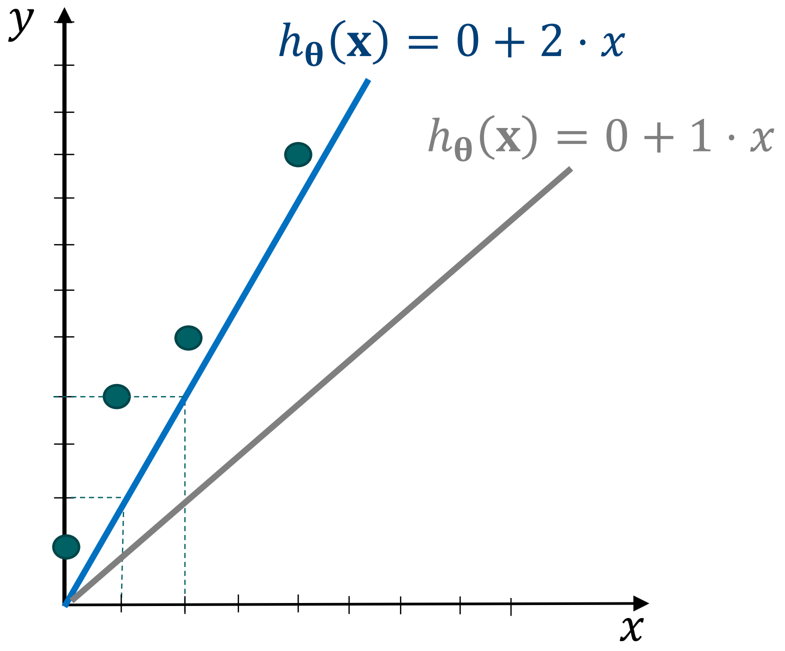


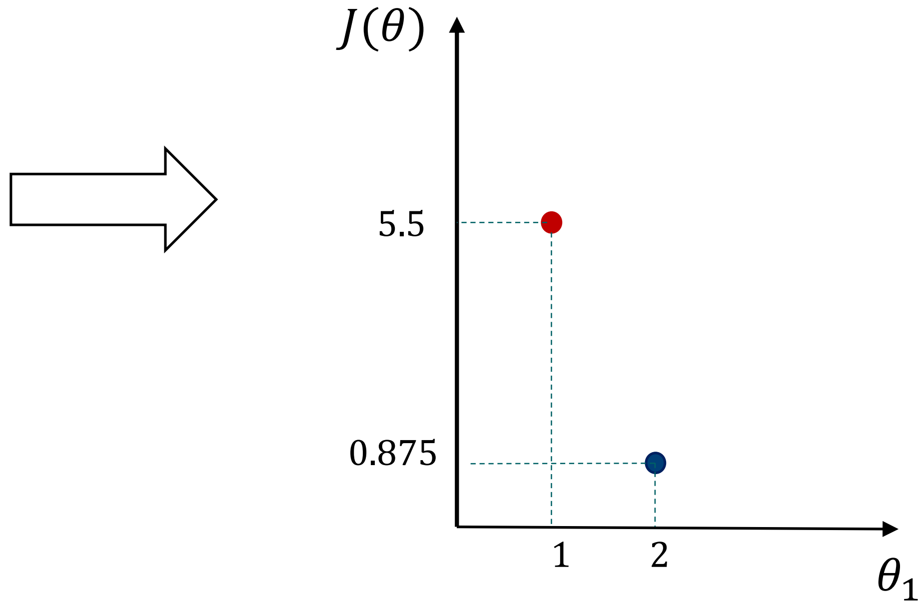
3’ Cost function computation $J(\theta_0 = 0, \theta_1 = 2)$

$$J(\boldsymbol{\theta}) = \frac{1}{2m}\sum_{i=1}^{m=4}\left[h_{\boldsymbol{\theta}}\left(x^{(i)}\right) - y^{(i)}\right]^2$$

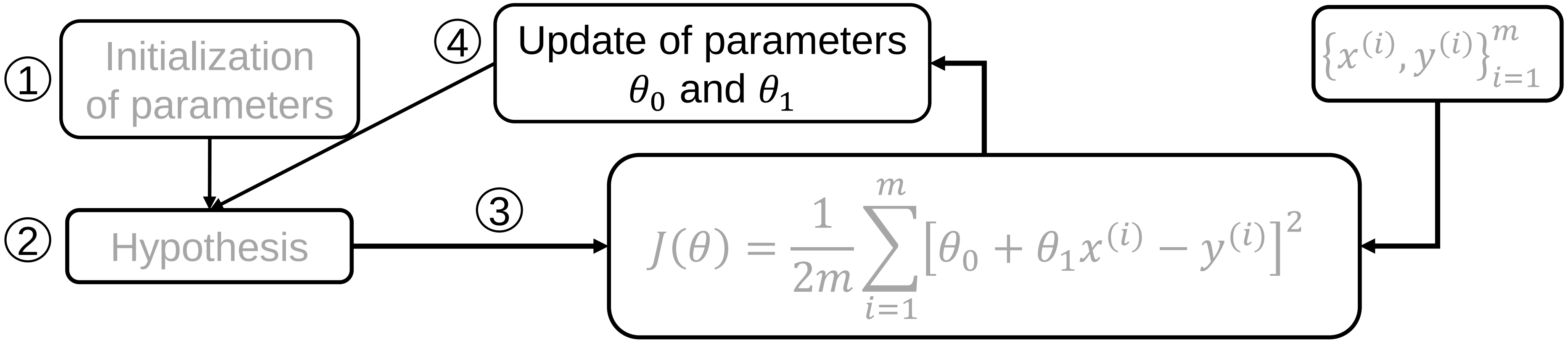
$$= \frac{1}{2m}\sum_{i=1}^{m=4}\left[0 + 2 \cdot x^{(i)} - y^{(i)}\right]^2$$

$$= \frac{1}{2 \times 4}\left([0-1]^2 + [2-4]^2 + [4-5]^2 + [8-9]^2\right) = 0.875$$





Linear Regression Training- 4' Parameter update



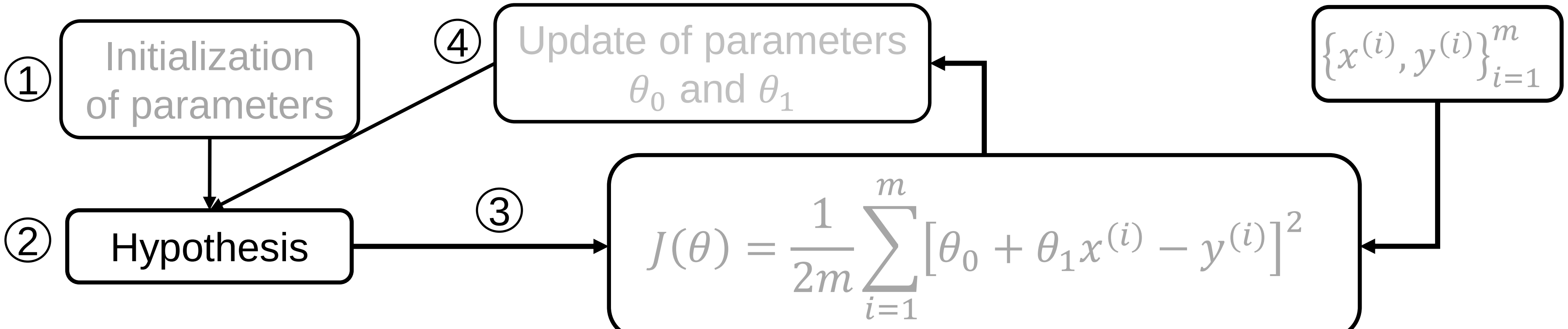
4' Parameters update for second iteration

- We update the parameters values:

$$\theta_0 = 0$$

$$\theta_1 = 3$$

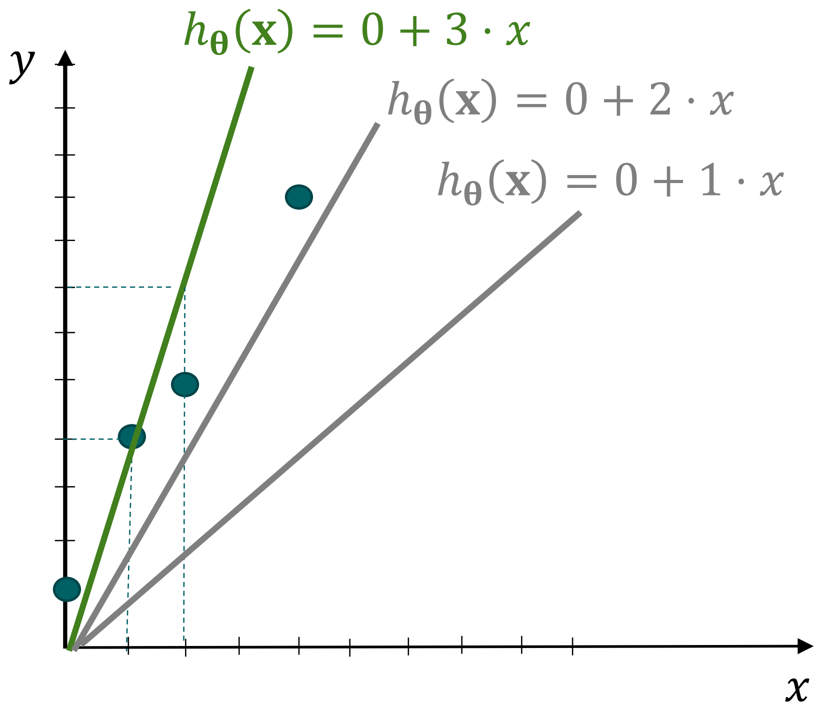
Linear Regression Training- 2'' Hypothesis



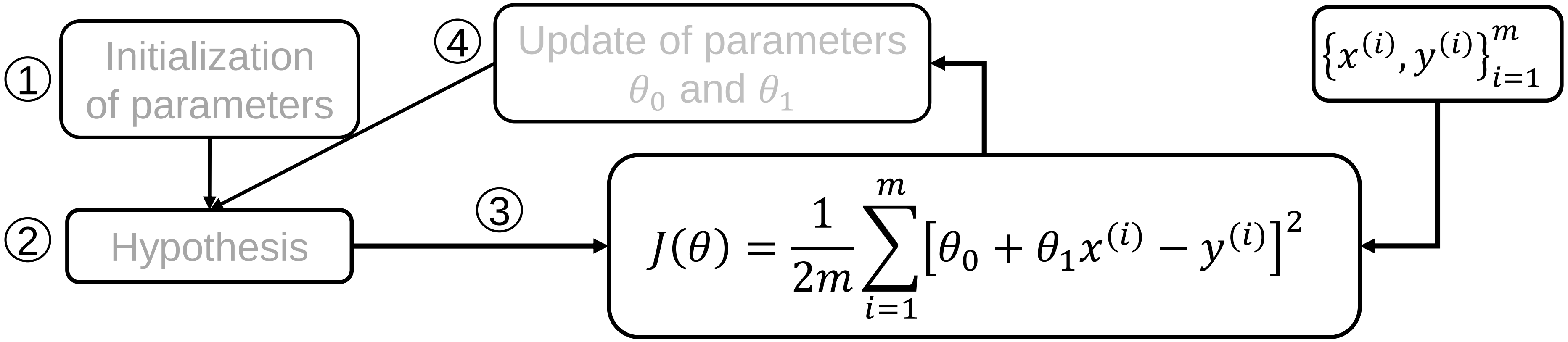
2'' Hypothesis for second iteration

- Then, for the given $\boldsymbol{\theta}$, the model is:

$$h_{\boldsymbol{\theta}}\left(x^{(i)}\right) = 0 \ + 3 \cdot x^{(i)}$$



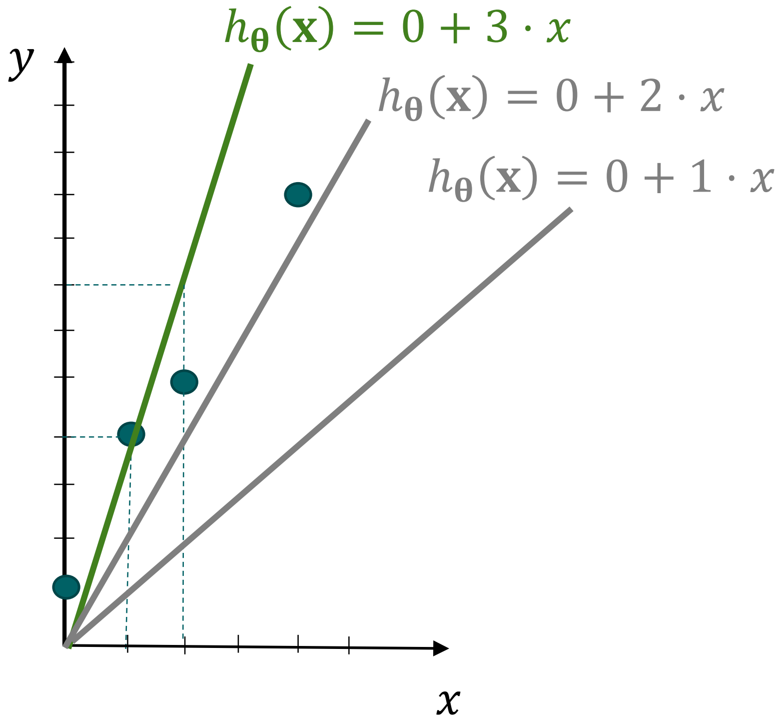
Linear Regression Training- 3'' Cost function

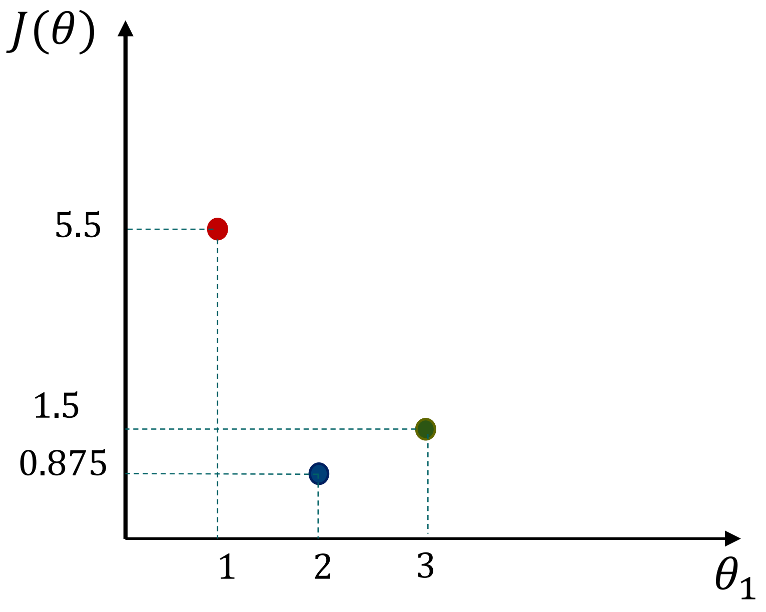


3'' Cost function calculation $J(\theta_0 = 0, \theta_1 = 3)$

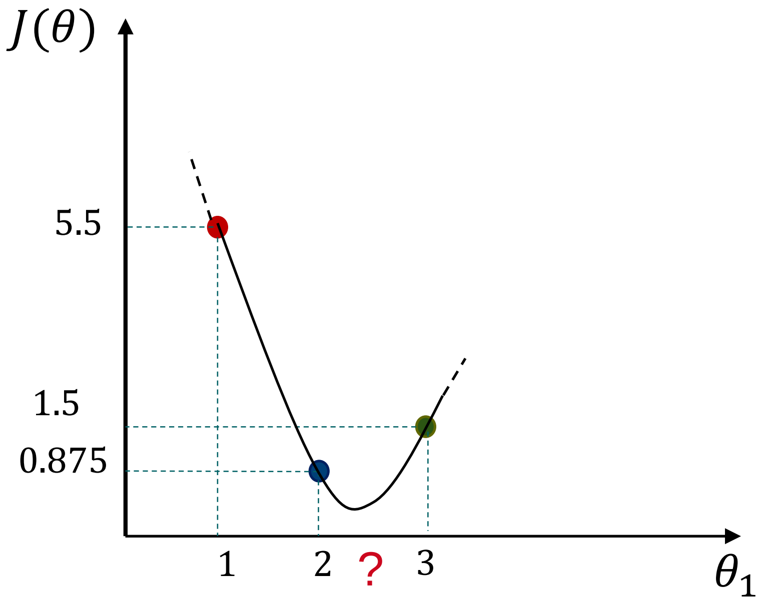
$$J(\theta) = \frac{1}{2 \times 4}([0-1]^2 + [3-4]^2 + [6-5]^2 + [12-9]^2)$$

$$= 1.5$$





What are the best parameters?



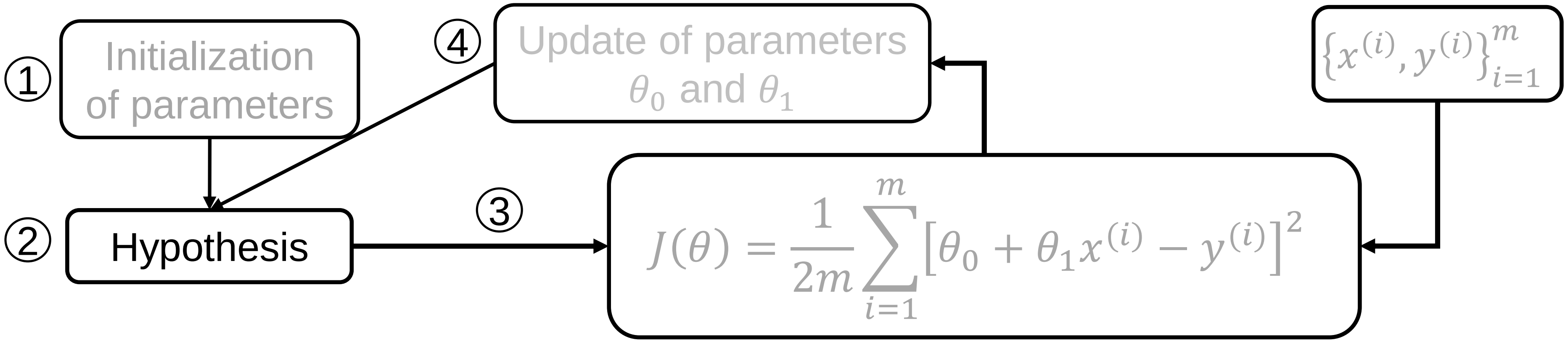
4'' Parameters update for third iteration

- We assume one value for θ_1 between 2 and 3. For example $\theta_1 = 2.5$
- We update the parameters values:

$$\theta_0 = 0$$

$$\theta_1 = 2.5$$

Linear Regression Training- 2''' Hypothesis

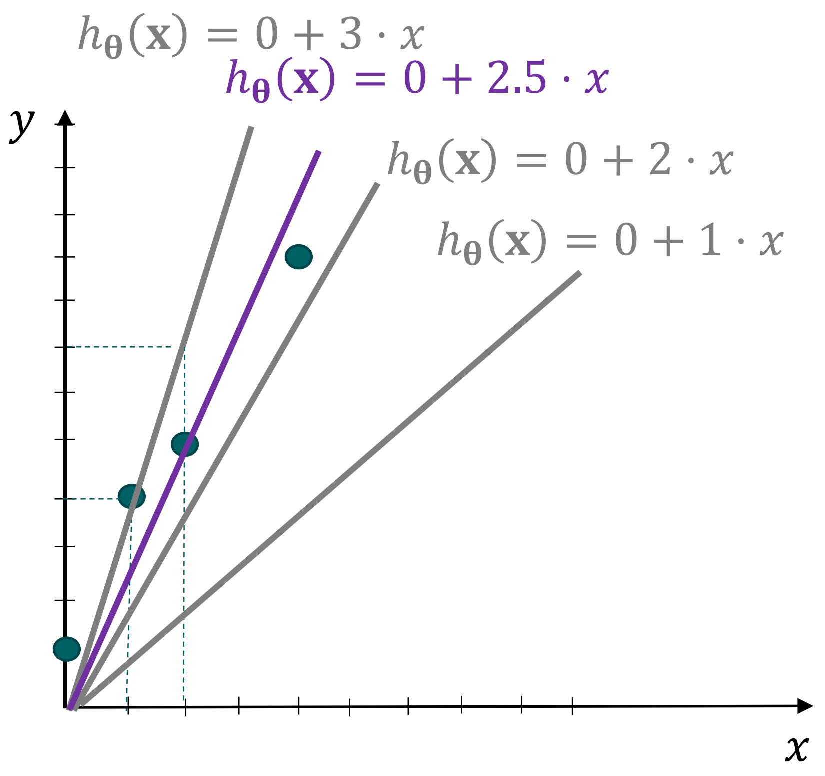


2'''

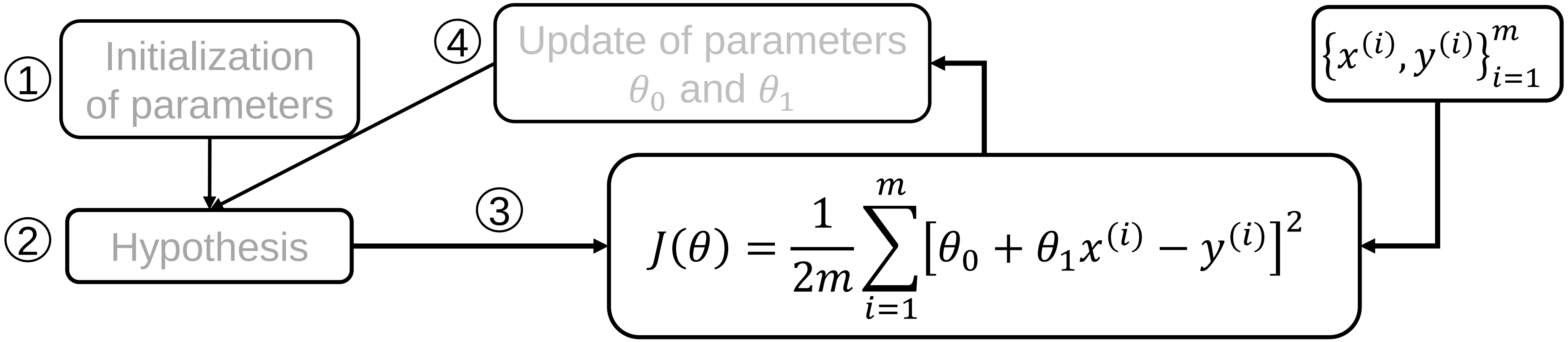
Hypothesis for third iteration

- Then, for the given $\boldsymbol{\theta}$, the model is:

$$h_{\boldsymbol{\theta}}\left(x^{(i)}\right) = 0 + 2.5 \cdot x^{(i)}$$



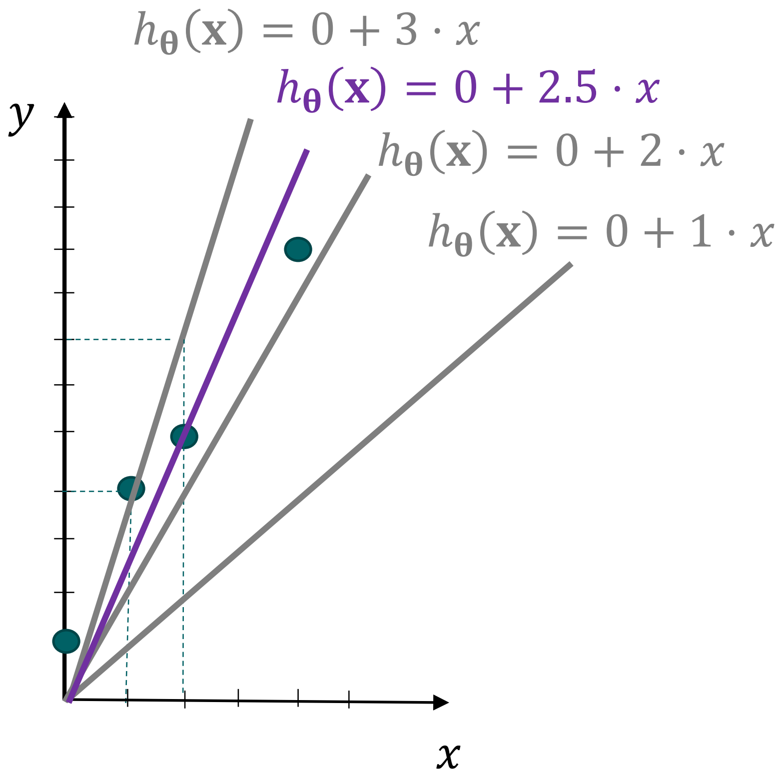
Linear Regression Training- 3''' Cost function

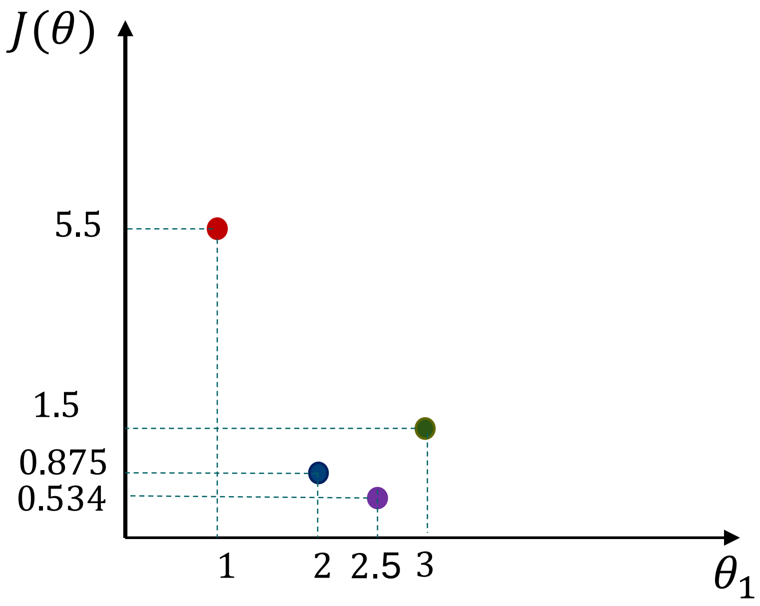


3''' Cost function calculation $J(\theta_0 = 0, \theta_1 = 2.5)$

$$J(\theta) = \frac{1}{2 \times 4}([0-1]^2 + [2.5-4]^2 + [5-5]^2 + [10-9]^2)$$

$$= 0.53$$





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Next Step

- Find the parameters which fits more to the dataset by applying same steps.(Grid search)

- Now check the performance on the testing set.

Assignment

Hypothesis model selection for linear regression using grid search

- **Objective:** Given a dataset $\{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$ generated from an underlying function $y(x)$ with added noise, find the best regression model (hypothesis) defined as $h_\theta(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$.

Tasks

Task1. Implement the linear regression model in Python as follows:

Step0. Randomly split the dataset into training and testing sets, using 25% of the data for testing

(Step1 is given in the code.)

Step2. Implement the hypothesis function indicated as *hypothesis(theta0, theta1, x)*

Step3. Compute the prediction error using the cost function indicated as

cost(y, yhat)

Step4. Perform a grid search to find the optimal values of (θ_0, θ_1) that minimize $J(\theta)$ indicated as *grid_search(x, y, theta0_values, theta1_values)*

Tasks

- **Task2.** Visualize the results by:
  - Plotting the $h_\theta(x)$ before and after updating the parameters (θ_0, θ_1)
  - Displaying a heatmap of the cost function $J(\theta)$ over the parameter grid

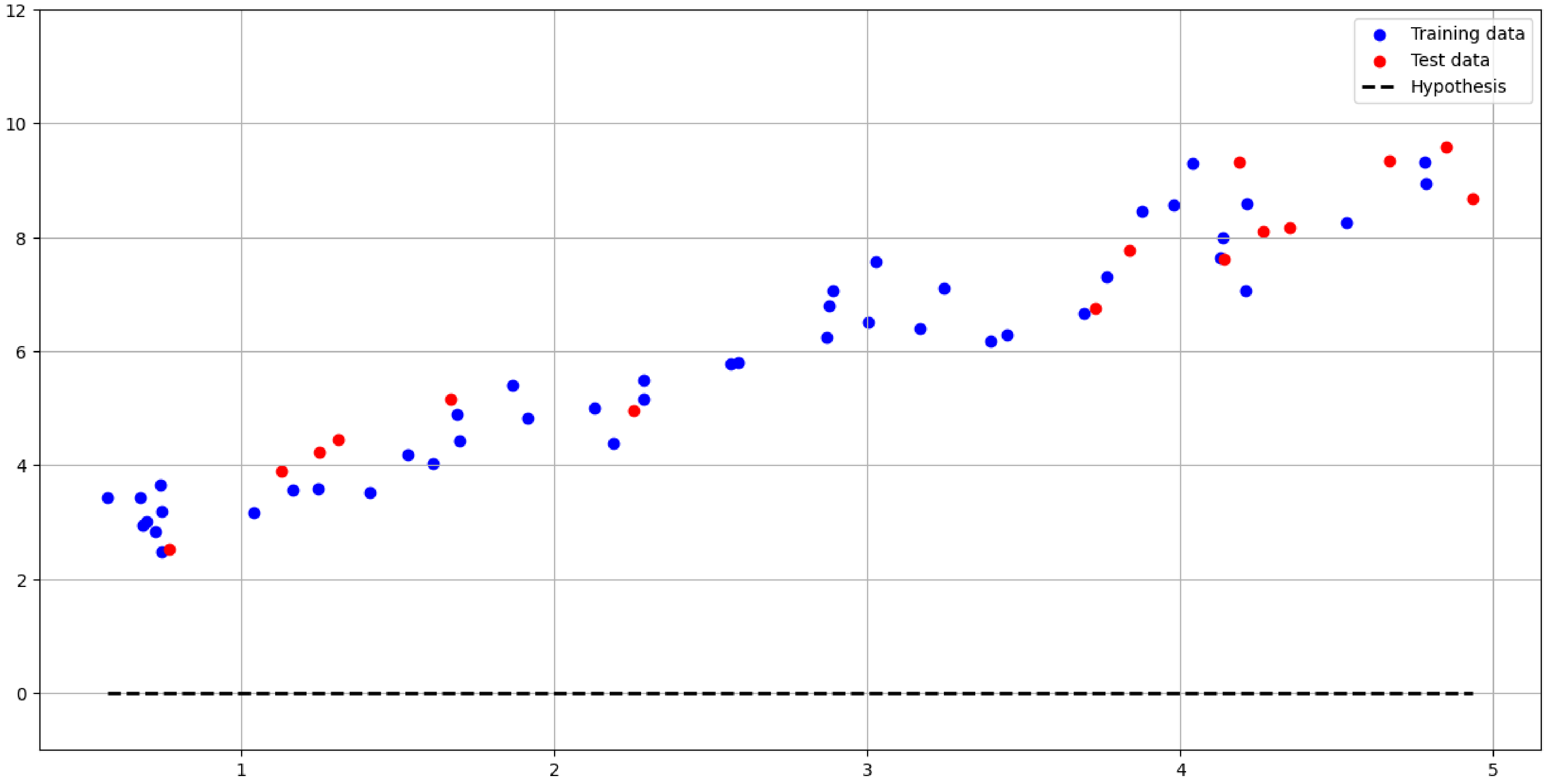
- Submit your code on Moodle by **May 11, 2026, at 10:30 AM**.

Submission

- Functions you need to implement are (Rest of the code is given):
  - *split_train_test(x, y, test_ratio=0.25, seed=123)*
  - *hypothesis(theta0, theta1, x)*
  - *cost(y, yhat)*
  - *grid_search(x, y, theta0_values, theta1_values)*
- Submit your Python code as .py file.
  - If you experience any issues uploading the $.py$ files directly, compress it into a .zip or .tar file and upload the compressed file instead
- Your code will produce the following outputs:
  - 2 plots showing the hypothesis function (before and after updating the parameters)
  - 1 heatmap plot illustrating the cost function values
  - Printing best parameters (θ_0, θ_1) and train/test cost values

❖ If the code fails to run, it will receive no points.

Expected results, Before training



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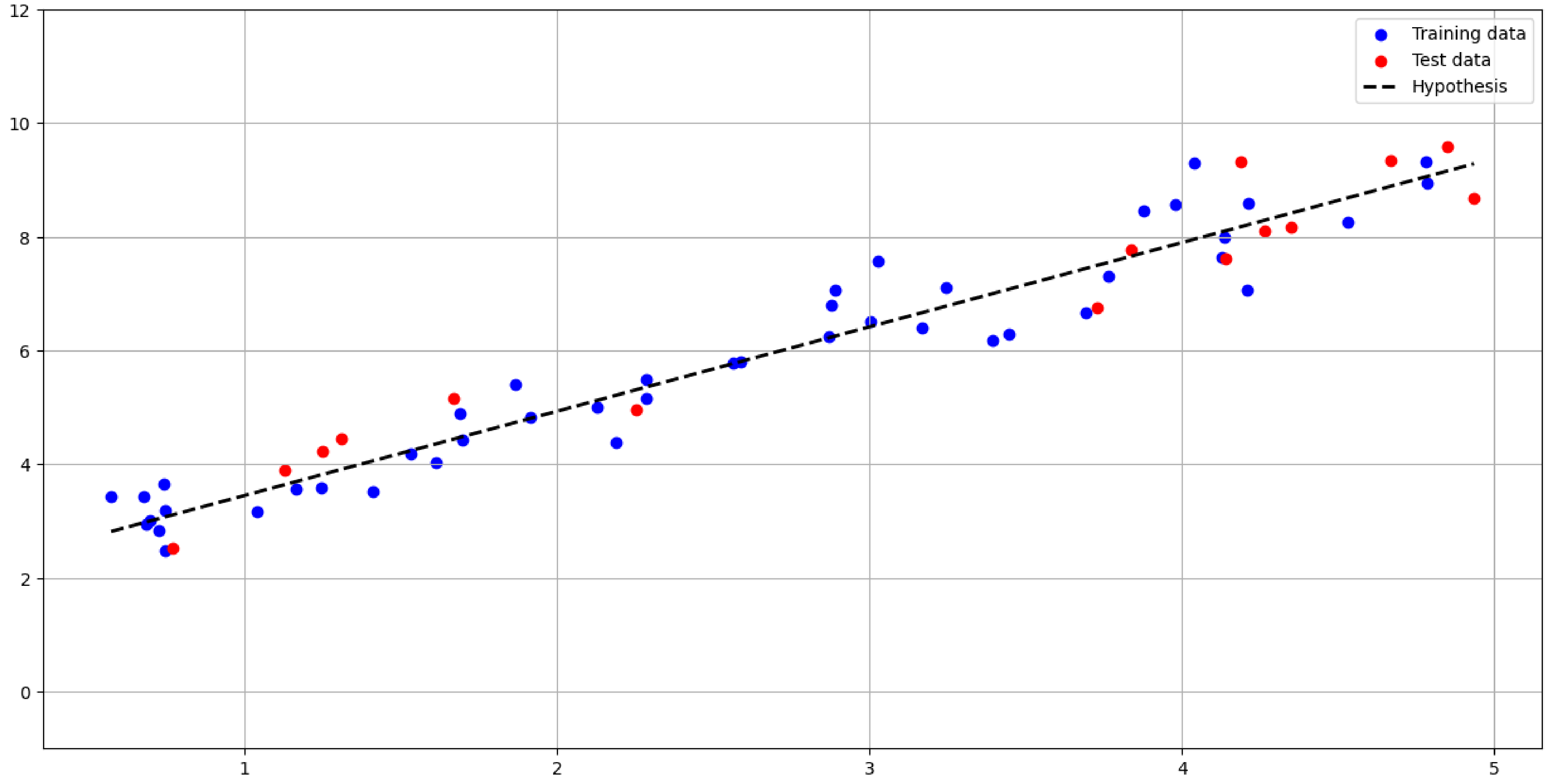
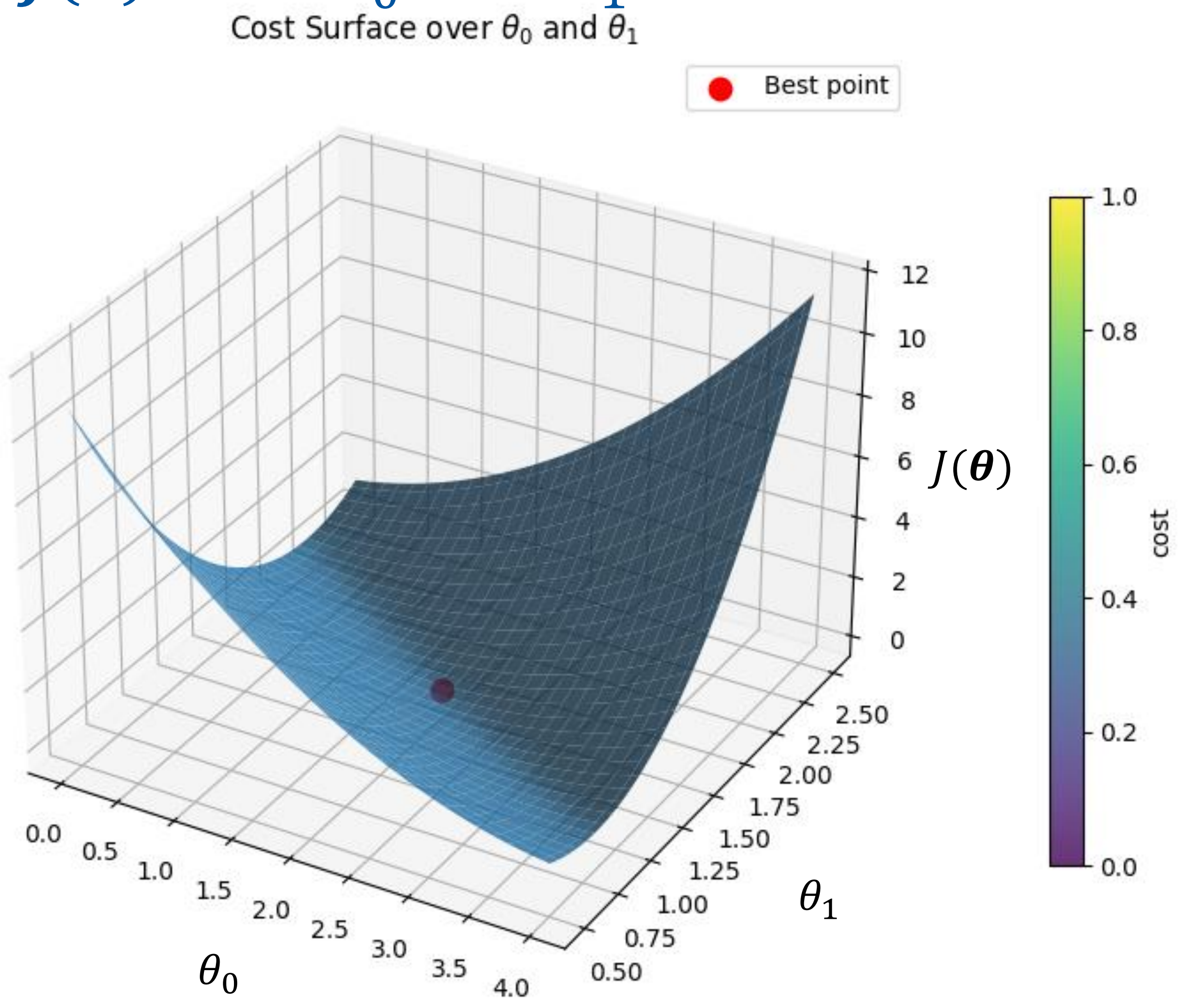
Expected results, After training



Expected results, $J(\boldsymbol{\theta})$ over θ_0 and θ_1



End of Exercise 2